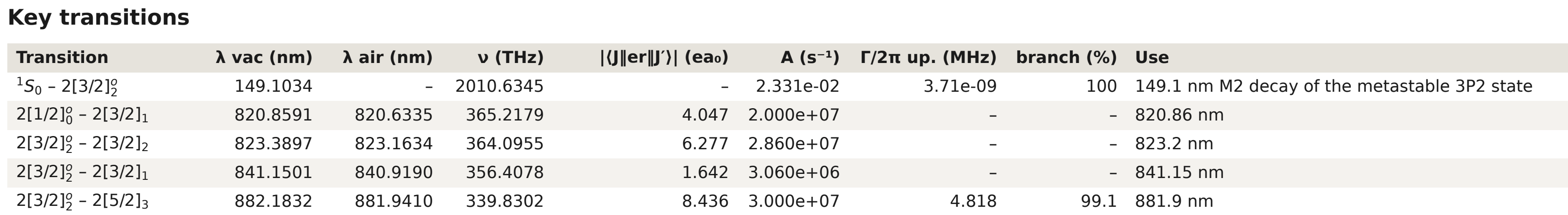
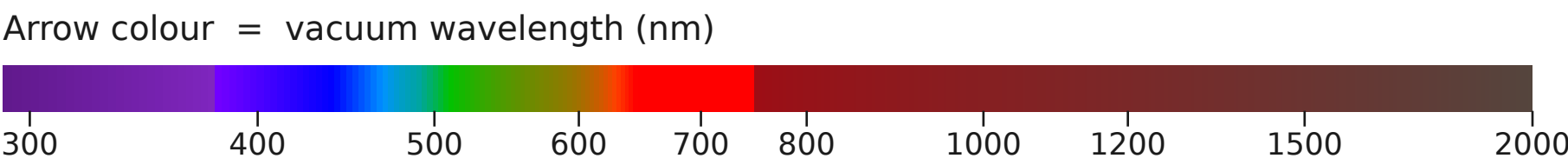


Xe (Xe I). Strongest classified transitions below 2 μm (NIST): 51 levels, 71 lines.

Xe⁺ ionisation limit 97833.79 cm⁻¹ = 12.12984 eV (102.214 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle f|er||i\rangle|$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: NIST energy and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

 ^{129}Xe hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₁	-2384.4(12)	-	F=3/2: +3576.60 F=5/2: -2384.40
2[3/2] ₂	-959.1(7)	-	F=1/2: +959.10 F=3/2: -479.55
2[1/2] ₂	-580.8(0.20)	-	F=1/2: +5808.00 F=3/2: -2904.00
2[3/2] ₁	-1308.9(6)	-	F=1/2: +1308.89 F=3/2: -654.45
2[3/2] ₂	-889.6(4)	-	F=3/2: +1334.40 F=5/2: -889.60
2[3/2] ₁	-4424.0(10)	-	F=1/2: +4424.00 F=3/2: -2212.00
2[3/2] ₂	-2895(5)	-	F=3/2: +4341.90 F=5/2: -2894.60
2[1/2] ₂	1977.0(0.10)	-	F=1/2: -1977.00 F=3/2: +988.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac. (nm)	Isotopes	Shift (MHz)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-134	-89.9(2)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-132	-132.4(21)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-131	-203(4)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-130	-182(5)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-129	-240(3)
$21/12\frac{1}{2}^- - 21/32_1^-$	820.859	136-128	-242.4(24)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-134	-95(4)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-132	-147(4)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-131	-222(7)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-130	-200(4)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-129	-275(5)
$21/32\frac{1}{2}^- - 21/32_1^-$	841.150	136-128	-261(5)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$